

Optimisation Methods to Address Variational Bayesian Posterior Variance Collapse

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Abstract

Variational Inference (VI) recasts Bayesian posterior computation as an optimisation problem, making it a natural testbed for comparing the gradient-based methods taught in ENEL445. This document provides the mathematical derivations for VI applied to Bayesian linear regression under a mean-field Normal–Gamma approximation.

We derive the Evidence Lower Bound (ELBO) in closed form and present the analytical Coordinate Ascent VI (CAVI) solution as a baseline. For gradient-based methods—gradient ascent, Newton’s method, and BFGS—we derive full gradient and Hessian expressions for all variational parameters $(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$, and formulate a quadratic penalty method to enforce minimum-variance constraints that prevent posterior variance collapse under the zero-forcing KL divergence.

The entry conditions are stated explicitly for each method. For line-search methods, a further condition must hold before each step is accepted: the search direction $\mathbf{d}^{(t)}$ must satisfy $\nabla \text{ELBO}(\boldsymbol{\phi}^{(t)})^T \mathbf{d}^{(t)} > 0$ (ascent direction), ensuring the line search is well-posed.

The step size $\alpha^{(t)}$ is determined by backtracking line search. Gradient ascent uses the Armijo sufficient-increase condition. BFGS uses the stronger Wolfe conditions—sufficient increase plus a curvature condition—to guarantee the gradient-difference vector $\mathbf{y}_t^T \mathbf{s}_t > 0$, which is required for the BFGS update to maintain positive definiteness of the approximate inverse Hessian.

Exit conditions are defined uniformly across all methods. Iteration terminates when any of the following hold: (i) gradient norm $\|\nabla \text{ELBO}\| < \varepsilon_g$; (ii) relative ELBO change $|\Delta \text{ELBO}|/|\text{ELBO}| < \varepsilon_r$; (iii) parameter change $\|\Delta \boldsymbol{\phi}\| < \varepsilon_\phi$; or (iv) a maximum iteration count is reached. CAVI additionally monitors $\max_i |\phi_i^{(t+1)} - \phi_i^{(t)}|/|\phi_i^{(t+1)}| < \varepsilon$.

Convergence rates and per-iteration costs are compared across all methods. BFGS offers the best practical balance of convergence speed and computational cost; CAVI remains competitive when conjugate structure can be exploited analytically.

Introduction

Bayesian inference provides principled uncertainty quantification but requires computing intractable posterior distributions. For most real-world models, the posterior integral,

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{\int p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}},$$

cannot be evaluated analytically. Variational Inference (VI) converts this inference problem into an optimisation problem, making it directly amenable to the gradient-based methods in this course.

Instead of computing the true posterior, VI finds the “best” approximation $q(\boldsymbol{\theta})$ from a tractable family $Q(\boldsymbol{\theta})$ by maximising the Evidence Lower Bound (ELBO). This transforms intractable integration into tractable optimisation—a natural testbed for the gradient-based methods.

A well-known failure mode of mean-field VI is *Posterior Variance Collapse*: the optimised approximation $q(\boldsymbol{\theta})$ is systematically overconfident, underestimating posterior uncertainty. This is a direct consequence of the KL direction minimised during ELBO maximisation, compounded by the mean-field factorisation assumption.¹

This paper addresses variance collapse via two complementary strategies,

- The first applies new optimisation methods to the *standard* ELBO objective: gradient ascent, Newton’s method, and BFGS are compared to determine whether the choice of optimiser affects dispersion quality.
- The second modifies the objective function itself, augmenting the ELBO with an entropy regularisation term $\lambda H[q(\boldsymbol{\beta})]$ to create a new objective $\mathcal{L}_{\text{reg}}(\phi) = \mathcal{L}(\phi) + \lambda H[q(\boldsymbol{\beta})]$ that explicitly incentivises broader posteriors.

Both the optimiser-comparison and modified-ELBO strategies are explored.

Throughout all gradient-based methods, the variational covariance $\boldsymbol{\Sigma}_{\beta}$ must remain symmetric positive definite at every iteration.² Standard gradient steps in Euclidean space can violate this requirement, leading to numerical failure. The solution is to parameterise the optimisation over the Cholesky factor $\mathbf{L}$ rather than $\boldsymbol{\Sigma}_{\beta}$ directly. Cholesky reparameterisation is a numerical implementation technique. Cholesky is the domain-specific choice for satisfying the constraint $\boldsymbol{\Sigma}_{\beta} \succ 0$ when applying those methods to this problem. The set of symmetric positive definite matrices is not a flat subspace of $\mathbb{R}^{p \times p}$: it has a curved boundary, and a straight-line gradient step can cross outside it, producing an invalid covariance matrix. Cholesky reparameterisation handles this by converting the constraint into a smooth bijection between the valid set and unconstrained $\mathbb{R}^{p(p+1)/2}$, so standard gradient steps always remain valid. It is standard practice in variational inference and Bayesian computation.

¹See ?? for the full mechanistic explanation.

²This structural constraint is satisfied automatically via Cholesky reparameterisation; see Appendix A. For the role of symmetric positive definite covariance matrices in exponential families more broadly, see Wainwright & Jordan (2008), §3.4, DOI: 10.1561/22000000001.

Key Notation

Symbol	Meaning
$\boldsymbol{\beta}$	Regression coefficient vector
$\mathbf{u}$	Random effects vector (hierarchical model)
$\boldsymbol{\varepsilon}$	Observation noise vector
τ_e	Observation-noise precision
τ_u	Random-effects precision
$\boldsymbol{\mu}_\beta$	Mean of variational posterior $q(\boldsymbol{\beta})$
$\boldsymbol{\Sigma}_\beta$	Covariance of variational posterior $q(\boldsymbol{\beta})$
$\mathbf{L}$	Lower-triangular Cholesky factor of $\boldsymbol{\Sigma}_\beta$
η_a, η_b	Log-space parameters for a_e, b_e

Optimisation Problem Formulation

Having established that VI converts posterior inference into optimisation, the problem must be stated precisely before any method can be applied. This section identifies the decision variables, the objective function, and the constraints, following the standard form used throughout the course. A well-posed formulation is necessary both for implementation and for meaningful comparison across methods: gradient ascent, Newton's method, and BFGS all operate on the same objective and respect the same constraints, so differences in performance reflect the methods themselves rather than problem differences.

Decision Variables

For Bayesian linear regression with p predictors, the variational parameters (decision variables) are:

- $\boldsymbol{\mu}_\beta \in \mathbb{R}^p$: posterior mean of regression coefficients
- $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{p \times p}$: posterior covariance (symmetric positive definite³)
- $a_e, b_e > 0$: shape and rate parameters of the variational Gamma factor for precision

Here, a_e and b_e parameterise $q(\tau_e) = \text{Gamma}(a_e, b_e)$; the model prior on τ_e is specified in the Test Case section.

Total dimension: $d = p + \frac{p(p+1)}{2} + 2 = 11$ variables for $p = 3$ predictors.

Objective Function

Maximise the Evidence Lower Bound (ELBO):

$$\text{ELBO}(\phi) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] + H(q) \quad (1)$$

$$= \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta}, \tau_e)]. \quad (2)$$

where $\phi = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ are the variational parameters.

Constraints

- $\boldsymbol{\Sigma}_\beta \succ 0$: Covariance matrix must be positive definite (see also: positive semidefinite, $\mathbf{A} \succeq 0$ ⁴)
- $a_e, b_e > 0$: Gamma parameters must be positive
- Optional: Minimum variance constraints $\sigma_{\beta_j}^2 \geq \sigma_{\min}^2$ to prevent underdispersion

Constraint Handling via Reparameterisation:

To enforce $\boldsymbol{\Sigma}_\beta \succ 0$, use Cholesky decomposition:

$$\boldsymbol{\Sigma}_\beta = \mathbf{L}\mathbf{L}^T \quad (3)$$

³A matrix $\mathbf{A}$ is **positive definite** if $\mathbf{v}^T \mathbf{A} \mathbf{v} > 0$ for every non-zero vector $\mathbf{v}$. Intuitively: the matrix amplifies every direction — there is no direction in which it squashes things to zero or flips them negative. For a covariance matrix this means every direction in parameter space has genuine positive uncertainty. Written $\mathbf{A} \succ 0$.

⁴**Positive semidefinite** ($\mathbf{A} \succeq 0$) allows $\mathbf{v}^T \mathbf{A} \mathbf{v} = 0$ for some non-zero $\mathbf{v}$, meaning some combination of coefficients has *exactly* zero uncertainty — the distribution is flat in that direction. A valid covariance matrix only needs to be positive semidefinite, but a semidefinite (not strictly definite) matrix is non-invertible⁵ and leads to degenerate directions.⁶

where $\mathbf{L} \in \mathbb{R}^{p \times p}$ is lower triangular with positive diagonal entries. The unconstrained parameters are:

- $L_{ii} = e^{\ell_{ii}}$ for $i = 1, \dots, p$ (exponentiate to ensure positivity)
- $L_{ij} = \ell_{ij}$ for $i > j$ (off-diagonal entries unconstrained)

Total Cholesky parameters: $\frac{p(p+1)}{2}$ elements of ℓ .

To enforce $a_e, b_e > 0$, use log-space parameterisation:

$$a_e = e^{\eta_a}, \quad b_e = e^{\eta_b} \quad (4)$$

where $\eta_a, \eta_b \in \mathbb{R}$ are unconstrained.

Unconstrained parameter vector: $\boldsymbol{\xi} = (\boldsymbol{\mu}_\beta, \ell, \eta_a, \eta_b) \in \mathbb{R}^d$ with $d = p + \frac{p(p+1)}{2} + 2 = 11$ for $p = 3$.

Computational Characteristics

- Gradient computation: $O(np^2)$ per evaluation (dominated by matrix operations)
- Hessian computation: $O(np^2 + d^3)$ for Newton's method
- Parameters exhibit coupling (changing $\boldsymbol{\mu}_\beta$ affects optimal $\boldsymbol{\Sigma}_\beta$)
- Must handle special functions: digamma $\psi(\cdot)$, trigamma $\psi'(\cdot)$ for Gamma distribution

Test Case: Bayesian Linear Regression

The test case uses Bayesian linear regression with:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \tau_e^{-1}\mathbf{I}_n) \quad (\text{likelihood}) \quad (5)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p) \quad (\text{prior on coefficients}) \quad (6)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e) \quad (\text{prior on precision}) \quad (7)$$

Mean-field approximation:

$$q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta}) q(\tau_e) \quad (8)$$

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \quad (9)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e) \quad (10)$$

Variational parameters: $\boldsymbol{\phi} = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ with dimension $d = p + \frac{p(p+1)}{2} + 2$

For $p = 3$ predictors: $d = 3 + 6 + 2 = 11$ parameters to optimise.

Overview of Optimisation Methods

Methods considered

- CAVI: Domain-specific baseline exploiting problem structure
- Gradient Ascent: First-order baseline method using gradient information and line search
- Newton's Method: Second-order method with quadratic local convergence near a well-conditioned optimum
- BFGS: Quasi-Newton method that approximates Hessian information; practical general-purpose method for smooth unconstrained problems

Baseline Method: CAVI⁷

Coordinate Ascent Variational Inference (CAVI) is VI-specific. It updates each variational parameter block sequentially using closed-form solutions:

$$\boldsymbol{\Sigma}_\beta \leftarrow \left(\frac{a_e}{b_e} \mathbf{X}^T \mathbf{X} + \lambda_\beta \mathbf{I}_p \right)^{-1} \quad (11)$$

$$\boldsymbol{\mu}_\beta \leftarrow \frac{a_e}{b_e} \boldsymbol{\Sigma}_\beta \mathbf{X}^T \mathbf{y} \quad (12)$$

$$a_e \leftarrow \alpha_e + \frac{n}{2} \quad (13)$$

$$b_e \leftarrow \gamma_e + \frac{1}{2} [\|\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}^T \mathbf{X} \boldsymbol{\Sigma}_\beta)] \quad (14)$$

Properties: Monotonic ELBO increase⁸, linear convergence, $O(np^2)$ per iteration. Provides analytical benchmark for validation.

⁷Blei, Kucukelbir, and McAuliffe (2017), Variational Inference: A Review for Statisticians (Journal of the American Statistical Association).

⁸Monotonic ELBO increase means the objective is non-decreasing across iterations: $\mathcal{L}^{(t+1)} \geq \mathcal{L}^{(t)}$. In CAVI, each coordinate update maximises the ELBO with respect to one variational factor while holding the others fixed, so each step cannot decrease the ELBO.